

PORTING-FROM-LAVA

Porting Lava autonomous-QA solutions → Boba-Base (submodules-driven)

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Maps the Lava deep-research playbook (Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md) onto **Boba-Base’s actual stack** (Go qBitTorrent-go + Python download-proxy/merge_service + plugins + frontend/extension + Jackett), respecting Boba-Base’s **TDD + anti-bluff (CONST-XII)** mandate. Both repos already share submodules/{containers, jackett, challenges, helixqa} + constitution, so the reusable pieces land **in the shared submodule** (consumed by both), and only thin glue stays per-repo.

Applicability triage (honest — not a blind copy)

Lava §	Solution	Applies to Boba-Base?		Why
§1	Containerized Android emulator	×	no	Boba-Base has no Android; UI is web (frontend/extension) → browser E2E (Playwright-in-container) is the analog, separate work
		×		
§2	Android guest-net ndc fix		no	Android-only
§7	Compose UI Challenge + off-main nav		no	Android-only

Lava §	Solution	Applies to Boba-Base?	Why
§0/§4	Egress-path insight + VPN-host egress	✓ YES — primary	Same datacenter-IP tracker block; affects Jackett + merge_service + download-proxy + plugin engines
§3	Configurable outbound proxy	✓ YES	download-proxy/qBitTorrent-go/Jackett tracker calls need a configurable egress
§5	Durable remote execution (systemd-linger)	▲ YES	run_all_challenges.sh/ci.sh/helixqa long runs on remote hosts get reaped by logind
§8	Jackett cookie-login + cookie-harvest	partial — already present	Boba-Base has nnmclub-cookie-*, extract-tracker-cookies.sh, extract-jackett-key.py; ADD the apikey-vs-cookie management distinction + test fake-equivalence
§6/§9	Anti-bluff QA verdict + diagnostics	YES	strengthen run_all_challenges.sh verdicts + the diagnostic gotchas

The four high-value ports (in order)

P1 — Durable remote execution (§5) [submodule: containers; glue: scripts/]

Problem Boba-Base shares: long QA/deploy runs on a remote host die when the SSH session ends. **Root cause:** remote systemd-logind KillUserProcesses reaps detached user processes (tmux/nohup/setsid all die). **Fix:** loginctl enable-linger + systemd-run --user --unit=<n> --collect bash runner.sh; poll systemctl --user is-active; sentinel file for completion. Avoid piping the long command through tail -N (buffers until exit) — log per-step artifacts. **Port:** add containers submodule helper pkg/remoteexec (or scripts/lib/durable-run.sh) shared by both repos; wire scripts/deploy-remote.sh + run_all_challenges.sh to use it. **TDD:** a test that

launches a 60s sleeper via the helper, drops the SSH session, and asserts it's STILL running after (fails against the old nohup approach).

P2 — Egress decision + VPN-host routing (§0/§4) **[submodule: containers; glue: scripts/ + Jackett/ proxy env]**

Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, **not** Cloudflare → FlareSolverr can't fix). Affects Jackett indexer fetches + merge_service/download-proxy + plugin engines. **Diagnosis (port the script):** `curl https://api.ipify.org (host IP) + curl -o /dev/null -w %{http_code} https://<tracker>/ direct vs via a VPN-host SOCKS proxy`. Different egress IP + 200 via proxy ⇒ confirmed. **Fix:** route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel `ssh -D 127.0.0.1:1080 -N <vpnhost>` (use `--socks5-hostname` for remote DNS); point Jackett + download-proxy + qBitTorrent-go at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. **Port:** containers submodule pkg/egress (tunnel up/verify) + scripts/egress-via-vpn.sh glue; reuse Boba-Base's existing ensure-macos-tunnel.sh style. **TDD:** assert the via-proxy egress IP ≠ direct host IP AND a known-blocked tracker returns 200 via proxy.

P3 — Configurable outbound proxy in the services (§3)

download-proxy (Python): `httpx/requests` honor `HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY` env natively — add an explicit `BOBA_UPSTREAM_PROXY` config that sets these for tracker-bound clients, with loopback bypass (`NO_PROXY=127.0.0.1,localhost,jackett`). **qBitTorrent-go:** set `http.Transport.Proxy` (socks5 native, remote DNS) from a `BOBA_UPSTREAM_PROXY` env — mirror Lava internal/`httpx/proxy.go`. **Jackett:** has a built-in proxy setting (configure via its API/ServerConfig). **Deploy gotcha (port):** the env must be FORWARDED into the containers (`docker-compose.yml` env / the boba-ctl deploy) — Lava's bug was a missing allow-list entry. **Verify on distroless via podman inspect, not exec printenv.** **TDD:** a test with a local proxy asserts the service's tracker request traverses it; falsifiability: disable the wiring → test fails.

P4 — Jackett cookie-login hardening (§8) **[submodule: jackett integration + helixqa fakes]**

Add to Boba-Base's existing cookie infra: Jackett's management API (`/api/v2.0/indexers, indexer /config`) needs a dashboard session cookie (empty-password POST `/UI/Dashboard`)

— the apikey only authorizes Torznab /results+/caps. If boba-jackett/extract-jackett-key.py only uses the apikey, ListIndexers/config silently 302→/UI/Login. **Critical for anti-bluff:** the test fake MUST 302-without-cookie like real Jackett (behavioral equivalence) or the gap stays hidden. **TDD:** fake 302s management without the cookie; assert discovery succeeds via the cookie path; falsifiability: remove cookie login → 302 failure.

Submodules-driven principle (per Decoupled Reusable Architecture)

- **containers submodule** is the home for P1 (durable-exec) + P2 (egress/tunnel) — generic, both repos + future projects consume. Contribute upstream first, then bump the pin in Lava AND Boba-Base.
- **jackett integration + helixqa** is the home for P4's cookie-login + the behaviorally-equivalent fake.
- **Per-repo glue only:** P3's env wiring (each service's HTTP client + each repo's compose/deploy).

Execution order (TDD per CONST-XII — each step RED→GREEN with pasted evidence)

1. P1 durable-exec helper in containers + wire
run_all_challenges.sh/deploy-remote.sh.
2. P2 egress diagnosis + VPN-host tunnel helper in containers +
scripts/egress-via-vpn.sh.
3. P3 BOBA_UPSTREAM_PROXY in download-proxy + qBitTorrent-go +
Jackett + compose env-forward.
4. P4 Jackett cookie-login + fake-equivalence test.
5. Re-run run_all_challenges.sh from the VPN-routed host → real
tracker search→download evidence.

Source playbook (full root-cause detail + Lava code refs):
Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md. Each P-item
there has the exact mechanism + the falsifiability rehearsal
to replicate under Boba-Base's CONST-XII.